

KULBIR SINGH AHLUWALIA

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SUMMARY

Fourth-year PhD candidate in the Department of Computer Science at the University of Illinois at Urbana-Champaign, advised by Prof. Julia Hockenmaier and Prof. Girish Chowdhary. Research focuses on natural-language grounding for outdoor field robots for navigation, recovery from navigation failures, and adaptive robot learning. Brings **excellent communication skills**—four semesters as a UIUC Teaching Assistant (CS498GC, CS444, CS519), co-developed a graduate mobile-robotics curriculum, authored public course and research websites, four peer-reviewed publications (ICRA 2025, IEEE TASE, Agronomy) and a featured article in *Optics & Photonics News*—and **high levels of creativity and problem-solving**, recognized by the National Bal Shree Award for Creative Scientific Innovation and multiple first-prize robotics/hardware awards.

EDUCATION

Ph.D. in Computer Science, University of Illinois at Urbana-Champaign, USA (Jun 2022–Present), GPA: 3.91/4

M.Eng. in Robotics, University of Maryland, College Park, USA (Aug 2019–May 2021), GPA: 3.88/4

B.Tech. in Electrical Engineering, Punjab Engineering College, India (Aug 2015–May 2019), GPA: 8.12/10

RESEARCH PAPERS

- **Ahluwalia, K.S.***; Jain, C.* WaypointGen: Learning Grounded Waypoint based Navigation for Mobile Manipulators. Coordinated Science Laboratory Student Conference (CSL SC) 2026, Machine Learning, Generative AI, and Signal Processing Session.
- Cuaran, J.; **Ahluwalia, K.S.**; Koe, K.; Uppalapati, N.K.; Chowdhary, G. Active Semantic Mapping with Mobile Manipulator in Horticultural Environments. (Accepted to ICRA 2025)
- Rangwala, M.; Liu, J.; **Ahluwalia, K.S.**; Ghajar, S.; Dhami, H.S.; Tracy, B.F.; Tokekar, P.; Williams, R.K. DeepPaSTL: Spatio-Temporal Deep Learning Methods for Predicting Long-Term Pasture Terrains Using Synthetic Datasets. Agronomy 2021, 11, 2245 (Special Issue: AI and Agricultural Robots).
- Liu, J.; Rangwala, M.; **Ahluwalia, K.S.**; Ghajar, S.; Dhami, H.S.; Tracy, B.F.; Tokekar, P.; Williams, R.K. “Intermittent Deployment for Large-Scale Multi-Robot Forage Perception: Data Synthesis, Prediction, and Planning”, 2021 (IEEE TASE, Transactions on Automation Science and Engineering).

AWARDS & HONORS

- **National Bal Shree Award for Creative Scientific Innovations** — Ministry of Human Resource Development, Government of India (2011): national honor for creativity and original scientific problem-solving.
- **First Prize**, Texas Instruments Hardware Hackathon (2017) — “Smart Garden” autonomous system.
- **First Prize**, B.Tech. Major Project (2019) — teleoperated gesture-controlled robotic arm.
- **First Prize**, Autonomous “Pick-n-Place” Transporter Bot, IIT Roorkee; **6th of 400 teams**, IIT Bombay Techfest.

EXPERIENCE

Earthsense Inc., Urbana, IL, USA

May 2025–Aug 2025

Position: AI Intern (Supervisor: Michael McGuire, Lead CV Engineer)

- Developed a natural-language-conditioned waypoint generation pipeline for 2D and 3D motion planners for enhancing outdoor robot navigation.
- Created an automatic labeling pipeline for large outdoor robot navigation datasets using Grounded SAM2, streamlining data processing.
- Deployed and benchmarked six open-source VLMs (Molmo-7B-D, Gemma-3-27B, Qwen-2.5-VL-72B, Qwen3-30B-A3B, Llama4-Scout, Spatial-VLM; 3B–72B params) for robot reasoning in image space and open-world NL-conditioned QA for 4-wheeled skid-steer outdoor robots.
- Enhanced ROS-based systems for real-world agricultural applications, directly supporting Physical AI advancement.

Qualcomm Technologies, Inc., San Diego, CA, USA

June 8–Aug 21, 2026

Position: Product Management Intern, Compute Chipset (req 3078866; team under Sr. Director Mandar Deshpande)

- Conduct market/competitive analysis for on-device AI/VLM/generative AI features on Snapdragon X Elite/X2 Elite silicon (positioning vs. Apple M-series, Intel Lunar Lake, AMD Ryzen AI for AI-PC/Copilot+ workloads).
- Synthesize customer/partner/engineering feedback into prioritized product requirements and roadmap inputs.
- Technical evaluation of VLM/multimodal inference workloads on prototype/production silicon (power/thermal/latency/memory/acc trade-offs on Hexagon NPU).
- Cross-functional coordination with silicon/software teams (AIMET quantization, AI Hub Models) + marketing/BD; prepare internal briefings on edge AI trends.

- Proposed contributions: VLM competitive study on NPU (vs. Apple/Intel/AMD), on-device real-time VQA demo, model-quantization PRD (reusing EarthSense methodology on AIMET-grade INT4).

Teaching Assistant, CS498GC: Mobile Robotics for Computer Scientists, UIUC

Fall 2025

Instructor: Dr. Girish Chowdhary.

- Co-developed the graduate course curriculum (ROS 2, sensor fusion, SLAM, mobile manipulation); authored and maintain the public course website and assignments; hold weekly office hours and mentor students on robotics software.

Teaching Assistant, CS444: Deep Learning for Computer Vision, UIUC

Spring 2024, 2025, 2026

Instructor: Dr. Svetlana Lazebnik.

Teaching Assistant, CS519: Scientific Visualization, UIUC

Summer 2025

- Communicated complex data-visualization concepts to a multidisciplinary student audience and guided project work.

University of Illinois — Distributed Autonomous Systems Lab & Hockenmaier Lab

Aug 2022–present

Graduate Research Assistant. Mentors: Dr. Girish Chowdhary & Dr. Julia Hockenmaier.

- Built 3D, dynamic, semantic representations for objects and grounded 3D voxel value maps for outdoor in-the-wild robot navigation.
- Categorized and collected manual recovery action sequences for failure cases arising from visual occlusions, sensor noise, and mechanical failures.

TECHNICAL SKILLS

Languages	Python, MATLAB, C++
Analysis & Design	Simulink, Blender, AutoCAD, Solidworks, Ultimaker Cura, Adobe Premiere Pro
Libraries	PyTorch, Transformers, Numpy, Pandas, OpenCV, SciPy
Other	ROS, Gazebo, RViz, CUDA, Linux, Docker, Git, Doxygen, Anaconda, LaTeX

PROJECTS

- **WaypointGen: Grounded Waypoint Navigation for Mobile Manipulators** (CSL SC 2026; in prep IEEE RA-L): 14-step pipeline grounding free-form NL commands to 2D waypoints for outdoor mobile manipulators using Qwen-3-VL filtering, SLIC superpixel BEV decomposition, and MPPI trajectory selection. 6-shot SayPlan-style scaffolds; field deployment metrics.
- **Active Semantic Mapping with Mobile Manipulator** (ICRA 2025): Probabilistic semantic octomaps + NBV planning with information gain for orchard robots. Field deployment stats; core code contributions (with J. Cuaran et al.).
- **Enhancing Stereo Depth Maps via RGBD-Conditioned Diffusion** (CS598): Fine-tuned Stable Diffusion V2 for monocular depth; 46% Abs Rel improvement over Marigold/Depth-Anything-V2 on IRS/VKITTI/NYUv2.
- **EarthSense AI Intern projects** (2025): 6 VLMs (Molmo-7B-D to Spatial-VLM, 3B–72B) deployed on Jetson for outdoor robot NL reasoning + waypoint gen; SAM2 auto-labeling pipeline; ROS enhancements for ag robots.
- **Self-adjusting roadmaps (LD-PRM)**: Lifelong Dynamic PRM that restructures as occupancy evidence is gathered in unknown environments.
- **Optimized GestureGAN for resource-constrained settings**: MobileNet-backed cross-view image gen with $5.7\times$ param reduction.
- **Autonomous Vaccine Delivery Robot**: QR/arrow-based nav + vision/IMU/encoder/ultrasonic for block retrieval/transport in arena. 1st prize IIT Roorkee, 6th/400 at IIT Bombay; featured video.
- **Image segmentation using superpixels**: SLIC + VGG16 feature extraction pipeline for superpixel classification.
- **A* / Dijkstra Path Planning on Turtlebot3 / rigid robots**: Configuration-space A*/Dijkstra with non-holonomic constraints and obstacle avoidance (ENPM661).
- **Teleoperated gesture-controlled robotic arm**: Web-based remote control with live video + gesture for object transport between rooms; on-board power, LED, speaker (B.Tech major project, awards).
- **ARIAC 2019 industrial system**: UR10 arms + conveyors + AGVs for part picking, fault disposal, assembly, AGV delivery.